

MUHAMMAD HAMEED

AI Engineer

+92-301-1240398 · eng.hameed2943@gmail.com · Peshawar, Pakistan

Portfolio: [muhammad-hameed.vercel.app](#) · LinkedIn: [linkedin.com/in/muhammad-hameed-4803ba2b2](#) · GitHub: [github.com/muhammad-hameed-ai](#)

AI engineer specializing in computer vision, retrieval-augmented generation, and production ML systems. Built SkinGuard AI, an offline melanoma-screening system that quantified and closed a dermoscopy-to-smartphone domain gap (AUC 0.94 → 0.49 → 0.78) with calibrated, on-device inference, and an Android security system running real-time face recognition and weapon detection entirely on-device, backed by 397 automated tests.

EDUCATION

B.S. Software Engineering — CECOS University of IT & Emerging Sciences, Peshawar

July 2026

Coursework: Artificial Intelligence · Digital Image Processing · NLP · Database Systems · Data Structures & Algorithms

TECHNICAL SKILLS

Languages: Python, Kotlin, SQL

Computer Vision: TensorFlow, PyTorch, TensorFlow Lite, YOLOv8, FaceNet, EfficientNet, ONNX, Grad-CAM

RAG & LLMs: LangChain, Qdrant, BM25, HyDE, Cross-Encoder Reranking, FastAPI, Google ADK 2.0

Mobile & Android: Jetpack Compose, Clean Architecture, Room, CameraX

MLOps, Data & Tools: DVC, MLflow, Evidently AI, GitHub Actions CI/CD, JUnit, Turbine, Robolectric, Docker, Pandas, NumPy, Power BI, Git, Google Cloud Run

INTERNSHIP EXPERIENCE

Data & Systems Intern — Shaheen Arms Factory, Peshawar

2 Months · 2025

- Designed a structured data system to replace a fully manual, paper-based production log for a CNC manufacturing facility — tracking raw material intake, machine workloads, and finished-goods output so management could check production status directly instead of coordinating across departments.
- Turned that data into weekly reports for factory leadership, cutting the time needed to identify which machine was falling behind schedule.

Technology Intern — KP IT Board, Peshawar

2025

- Worked on web-based systems inside a government technology institution and learned how software requirements get scoped and approved in a public-sector environment — a slower, more document-driven process than commercial work.

PROJECTS

SecureVision — On-Device AI Security System (Android)

Aug 2026

Live: [muhammad-hameed-ai.github.io/securevision](#) · GitHub: [github.com/muhammad-hameed-ai/securevision](#)

Kotlin · Jetpack Compose · TensorFlow Lite · FaceNet-512 · YOLOv8 · JUnit · Turbine · Robolectric

- Engineered a fully offline security system with zero network access:** real-time face recognition (FaceNet-512, 89–98% match confidence) combined with a self-trained weapon detector (YOLOv8, mAP50 0.80), plus motion sensing, alarms, and event recording.
- Traced a false-rejection pattern on landscape captures to a stale 30° roll-tolerance threshold — rotating the phone shifts a standing subject's apparent roll by up to 90°, which the alignment step should have already corrected. **Logged per-face orientation data instead of guessing, then recalibrated to 45°.**
- Built to production standards rather than prototype standards: 20-module clean architecture, **397 JUnit unit tests** (Turbine for Flow testing, Robolectric for Android-runtime units) covering the face matcher, alignment transform, quality gate, and authentication layer, full CI/CD, and only 3 runtime permissions. Reliable recognition holds to 3–5 meters, the practical resolution limit of a phone camera sensor.

Saudi Vision 2030 Policy Intelligence Hub

Jul 2026

Live: [saudi-vision-2030-rag-3.onrender.com](#)

Python · FastAPI · Qdrant · LangChain · HyDE · DVC · MLflow · GitHub Actions

- Improved answer faithfulness from 0.10 to 0.52 (+420%)** by implementing HyDE retrieval to close the gap between short user queries and long-document embeddings, measured automatically in CI on every push.
- Indexed 51 government policy documents (2,184 pages, 6,257 vector chunks) through a hybrid pipeline — dense search fused with BM25 via Reciprocal Rank Fusion, reranked by a cross-encoder — backed by DVC-versioned experiments, MLflow tracking, and a CI gate that fails the build if faithfulness regresses. Deployed live at zero hosting cost.

SkinGuard AI — Offline Melanoma Screening (Final Year Project)

HEC-Registered · ICHARMM-2025 3rd Place

Live: [muhammad-hameed-ai.github.io/skinguard-ai](#) · GitHub: [github.com/muhammad-hameed-ai/skinguard-ai](#)

Python · PyTorch · ONNX · EfficientNet-B2 · MobileNetV3 · Flutter · Domain Adaptation

- Quantified a dermoscopy-to-smartphone domain gap most published melanoma classifiers never measure:** AUC collapsed from 0.9395 on dermoscopy to 0.4941 (chance) on phone photos, then recovered to 0.7778 through staged domain adaptation.
- Built a three-stage on-device pipeline — quality check, an out-of-distribution gate (MobileNetV3-Small, 99.41% accuracy) that refuses non-lesion images before either classifier runs, and dual diagnostic heads — then calibrated outputs via temperature scaling and isotonic regression, cutting expected calibration error from 0.2235 to **0.0072**.
- Audited 27,577 raw images for quality, duplication, and train/test leakage, removing 1,293 leaked and 3,594 duplicate images before reporting results. Documented two failed approaches along the way — INT8 quantization (0.16 AUC worse) and shared-checkpoint fine-tuning (0.056 AUC of catastrophic forgetting).

Google AI Agents Capstone — Kaggriculture

2026 · Google Developer Program × Kaggle

Google ADK 2.0 · Gemini · Model Context Protocol · Google Cloud Run

- Built a multi-agent system on Google's Agent Development Kit for a live-leaderboard simulation, with agents communicating over Model Context Protocol. Added prompt injection defenses and an LLM-as-a-Judge evaluation loop after testing exposed how easily unguarded agents could be steered off objective.

CERTIFICATIONS & AWARDS

- NASA Open Science 101** | NASA Science Mission Directorate · Aug 2026
- Google Advanced Data Analytics Professional Certificate** | Google / Coursera · Oct 2025 · ID: VKWOQT55EYDC
- Google 5-Day AI Agents · 1 Million Prompts (Dubai) · AI for Social Impact (ADBI)** | 2026
- Cisco Networking Academy Certificate** | Cisco · Feb 2026
- AWARD — 3rd Place, ICHARMM-2025 SE Poster Presentation** | CECOS University · 2025